Construction: Figure 6.18 shows the structure and symbol of DIAC. The DIAC is a two terminal device, namely MT_1, MT_2. It is a combination of parallel semiconductor layers ($P_1N_1P_2N_2$, $P_2N_1P_1N_3$) connected in anti-parallel. The DIAC can be configured to conduct in both the directions. The structure of DIAC is similar to transistor, but no terminal attached to the base layer.

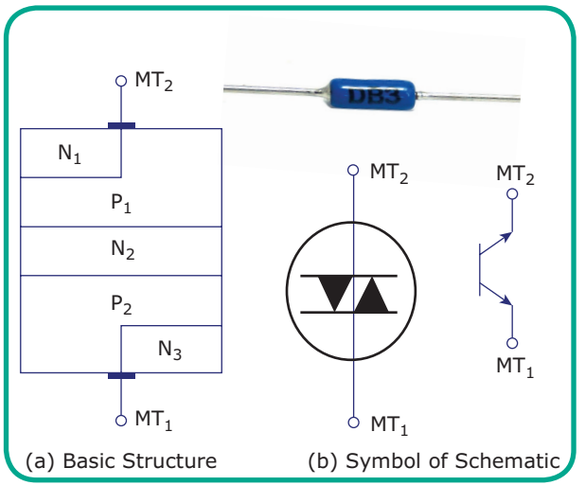


Figure 6.18 **Basic Structure and Symbol of DIAC**

Working Principle: The DIAC can conduct in either direction depending upon the polarity of the voltage applied across MT_2 and MT_1. It can be switched ON only, when the applied voltage reaches the break-over voltage (V_{BO}).

When MT_2 is positive with respect to MT_1, the junction J_1 and J_3 are forward biased, whereas junction J_2 andJ_4 are reverse biased. Therefore, no current flows through the diode $P_2N_1P_1N_3$. Since the junction J_2 is under reverse biased condition, initially a small leakage current flows through the diode $P_1N_1P_2N_2$. When the applied voltage exceeds the break over voltage, the reverse biased junction J_2 breaks due to avalanche effect. Therefore, the current flows from MT_2 to MT_1 through $P_1N_1P_2N_2$ as shown in Figure 6.19.

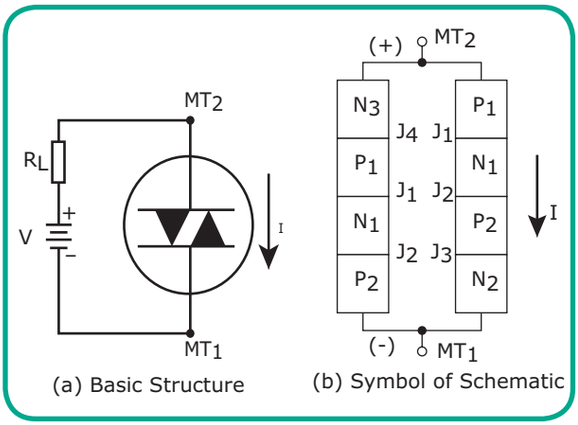


Figure 6.19 **Structure and Symbol Schematic when MT_2 is positive w.r.t. MT_1**

When MT_2 is negative with respect to MT_1 the junction J_2 and J_4 are forward biased, whereas junctions J_1 and J_3 are reverse biased. Therefore, no current flows through the diode $P_1N_1P_2N_2$. Since the junction J_1 is under reverse biased condition, initially a small leakage current flows through the diode $P_2N_1P_1N_3$. When the applied voltage exceeds the break over voltage, the reverse biased junction J_1 breaks due to avalanche effect.

Therefore, the diode passes current (I) through the diode $P_2N_1P_1N_3$ from MT_1 to MT_2 as shown in Figure 6.20.

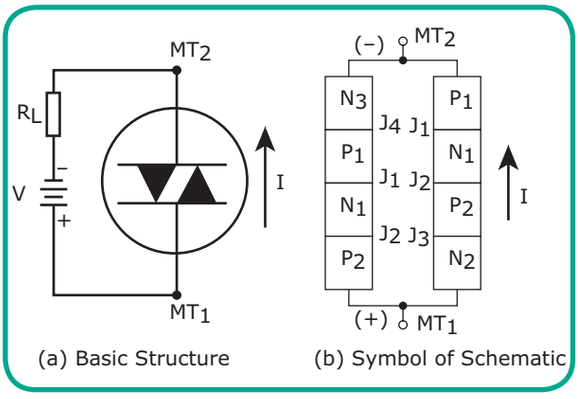


Figure 6.20 **Structure and Symbol Schematic when MT_2 is negative w.r.t. MT_1**

Application of DIAC

1. Used as Triggering device in TRIAC Power Control System.

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